



**BALL STATE
UNIVERSITY**

Module 4 Lecture - One-way ANOVA and Multiple Comparison Procedures

Analysis of Variance

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Table of Contents

1	Overview and Introduction	2
1.1	Objectives	2
1.2	Introduction	2
2	The F Distribution	3
2.1	Introduction	3
2.2	Additional Facts About the F Distribution	4
3	The F Ratio	5
3.1	Introduction	5
3.2	Between & Within Variation	6
3.3	Calculation of the F-ratio/statistic	6
3.3.1	Overall F Calculation	6
3.3.2	Between-group Calculations	6
3.3.3	Within-group Calculations	7
3.4	Conclusions with the F-ratio/statistic	8
4	One-way ANOVA	9
4.1	Introduction	9
4.2	Assumptions	9
4.3	Null and Alternative Hypotheses	10
4.4	Extensions	11
4.5	Effect Size for One-way ANOVA	12
4.5.1	Eta-squared	12
4.5.2	Omega-squared	13
5	Mean Comparisons in One-way ANOVA	13
5.1	Introduction	13
5.1.1	Types of Contrasts	14
5.1.2	Concept of Weights	14
6	Conclusion	15
6.1	Recap	15
	Key Terms	16

1 Overview and Introduction

1.1 Objectives

- Interpret the F probability distribution as the number of groups and the sample size change.
- Discuss one use for the F distribution: one-way ANOVA
- Conduct and interpret one-way ANOVA.
- Appreciate the ANOVA as another test in your toolbox alongside the other tests for dealing with categorical and continuous outcome variables
- Understand the F-distribution as another practical distribution for deriving inferences and conclusions
- Understand the types of effect size usually employed for ANOVA
- Introduce the different mean comparison strategies in a One-way ANOVA, to be followed up on later in next lecture

1.2 Introduction

 Discuss: When we needed to compare the averages of two groups on a continuous variable, what statistical test would we use?

- While we've already covered how to test _____ between two groups (even with an _____ violation!), now we need to consider the scenario with _____ or more groups on some outcome
 - Examples:
 - * Comparing class outcomes for those in hybrid, in-person, or online sections
 - * Comparing students on three different tracks in high school: regular, advanced, or remedial
- When we are comparing three _____ groups on some numeric, _____ outcome, we will employ the one-way ANOVA, standing for **Analysis of Variance (ANOVA)**
 - ANOVA is a _____ family of techniques with many applications and extensions - many of which we will cover in this class!

- We are looking at the one-way ANOVA as a _____ use of this technique

! Important

'ANOVA is a t-test, but for 3 categories' is actually a pretty fair way to describe this, as they are actually equivalent if/when comparing only two groups.

- Why not use a bunch of _____ across the 3 groups instead of this new complicated method?
 - Problem: It raises our Type I error rate
 - This is something done as part of ANOVA in _____ or _____ testing

🗨 Discuss: Describe what Type I errors are, in your own words

- Previous tests all made use of specific, practical _____ for determining significance:
 - z-tests → normal distribution
 - t-tests → t-distribution
 - χ^2 tests → χ^2 distribution
 - The ANOVA family will introduce a new distribution: the _____
- We'll start by describing the F-distribution itself, then the F-ratio/statistic that we compute, and then finally the _____ within the One-way ANOVA

2 The F Distribution

2.1 Introduction

- The **F-distribution** was developed by Sir Ronald Fisher and is a unique distribution applied often for the _____ family of statistical tests
 - In _____ it is given as $F \sim F_{df(num), df(denom)}$ where:
 - $df(num) \rightarrow df_{between}$ and $df(denom) \rightarrow df_{within}$

– Example: $F \sim F_{2,24}$ has:

* $df_{between} = 2$

* $df_{within} = 24$

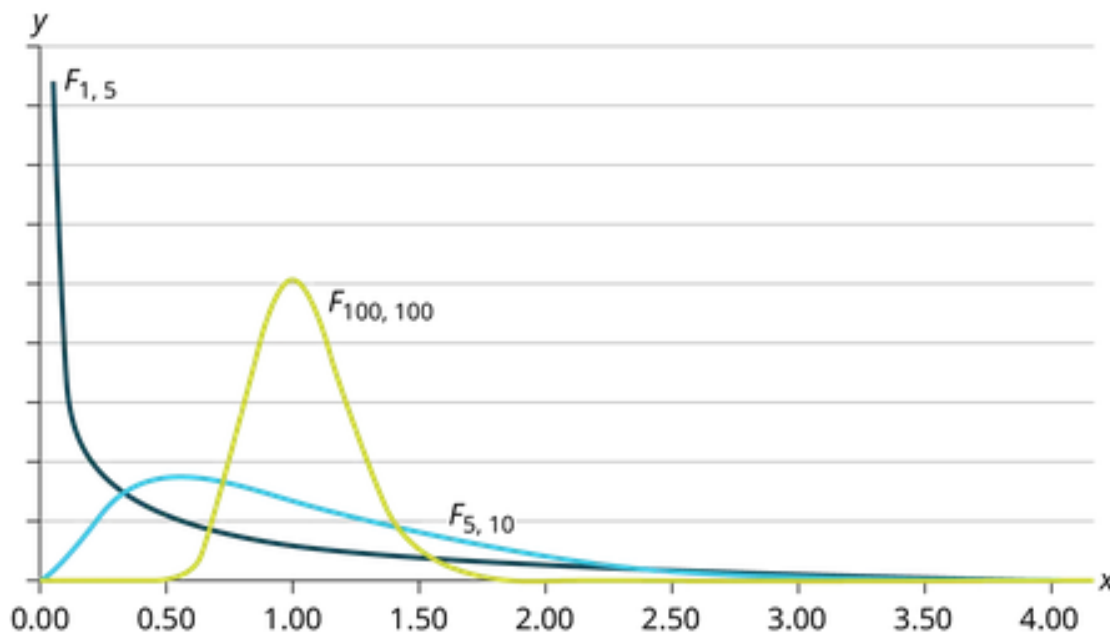
❗ Important

Hold on just a second, *two* different degrees of freedom? Yes! We'll discuss more about why that is in our discussion on the F-ratio itself

- The F-distribution actually consists of values that are _____ of the t-distribution, and is derived from the t-distribution
 - How exactly Fisher proved that is way beyond the scope of this class

🔊 Discuss: Recall, what exactly is the distinction that makes the t-distribution different and useful over the plain normal distribution? The F-statistic benefits from this the same!

2.2 Additional Facts About the F Distribution



"I don't mind not knowing. It doesn't scare me." — Richard P. Feynman

- The curve is not _____ but skewed to the right.
 - Thus, it has a more similar look to the _____ distribution, compared to the t-distribution
- There is a different curve for each set of _____ .
 - Once again, similar to chi-squared
- The F statistic is always greater than or equal to zero.

 Discuss: After you work through the F-ratio section and calculation below, return to this section and try to explain, using the formula, why this F is always greater than or equal to zero

- As the degrees of freedom for the numerator and for the denominator get larger, the curve approximates the _____ distribution.
 - Remember, the $df_{between}$ → numerator
 - The df_{within} → denominator

3 The F Ratio

3.1 Introduction

- The **F-ratio** is just another name for the F statistic that _____ from the test we run
 - The F-ratio/statistic we get is a ratio of:
 - * The variances _____ samples
 - * The variances _____ samples
 - * It's formula is given as:

$$F = \frac{MS_{between}}{MS_{within}}$$

- We'll describe what the MS means here in a second

3.2 Between & Within Variation

- For **Variances between samples**, we are considering how much _____ there is between the 3 or more groups we are comparing to one another
 - This may also be known as “variation due to _____” or “_____ variation”
 - This will be represented _____ as the $MS_{between}$ or the **mean square**
 - This is the _____ of the F-ratio formula
- In the case of **Variances within samples**, we examine how much variation there is within _____ of the groups we are comparing to one another
 - This may also be known as “variation due to _____” or “_____ variation”
 - This will be represented as MS_{within} in the following formula
 - This is the _____ of the F-ratio formula

3.3 Calculation of the F-ratio/statistic

- We are *not* going to calculate F by hand, as it is very time-consuming, albeit plenty possible
 - However, you should still follow along and make sure you can see the flow of information as we substitute numbers in these calculations

3.3.1 Overall F Calculation

$$F = \frac{MS_{between}}{MS_{within}}$$

3.3.2 Between-group Calculations

$$MS_{between} = \frac{SS_{between}}{df_{between}}$$

$$df_{between} = k - 1$$

$$SS_{between} = \sum \left[\frac{(s_j)^2}{n_j} \right] - \frac{(\sum s_j)^2}{n}$$

$$SS_{total} = \sum x^2 \cdot \frac{(\sum x)^2}{n}$$

• Where

- $SS_{between}$: Sum of squares between groups
- SS_{total} : Sum of squares total
- $df_{between}$: degrees of freedom between groups
- k : the number of groups
- n : total sample size
- n_j : the size of j^{th} group
- s_j : sum of values of j^{th} group

3.3.3 Within-group Calculations

$$MS_{within} = \frac{SS_{within}}{df_{within}}$$

$$df_{within} = n - k$$

$$SS_{within} = SS_{total} - SS_{between}$$

• Where

- SS_{within} : Sum of squares within groups
- $SS_{between}$: Sum of squares between groups
- SS_{total} : Sum of squares total
- df_{within} : degrees of freedom within groups
- n : total sample size

3.4 Conclusions with the F-ratio/statistic

 Discuss: Review: describe the general processes of hypothesis testing, starting with setting the null and alternative hypotheses and ending with a rejection or retaining of the null hypothesis. Try and use the words 'rare event' at some point in your explanation.

- The F statistic, which we derive from the formula for whatever test we are using, is actually best described as a _____ or fraction (we'll return to this ratio/statistic in a bit!)
 - Much like previous test statistics, we use this to determine if our results are _____ enough to describe our results as unlikely if the null hypothesis is _____
 - As usual, the F statistic has a corresponding _____ that tell us the probability of results due to chance under the null hypothesis

 What value do we compare p against to determine if we can reject the null hypothesis?

- ☒ A) Alpha
☐ B) Omega
☐ C) Confidence Level
☐ D) Beta

Explanation:

- For _____ testing of the F-ratio, our one-way ANOVA test will always be _____ -tailed due to the ratio nature of statistic, with larger numbers suggest greater variation between groups relative to the variation within groups!
 - Put another way, we are trying to see if it is _____ enough out on the right tail to say that it is significant!

 Discuss: Which other distribution also always had a right-tailed application? Explain why that is using the relevant formula for that statistic.

4 One-way ANOVA

4.1 Introduction

 Important

In the prior calculations, we covered the mathematical computation in a one-way ANOVA, but this next part will cover the more conceptual side of things.

- The goal of the one-way ANOVA is to determine if there are _____ differences between _____ group means
 - This is done by examining the _____ of the groups (as we previously discussed as part of [The F Ratio](#))
- This is how we will practically use the F-distribution

4.2 Assumptions

- Much like the other tests, we need to be mindful of several _____ that underlie this statistical test
- These assumptions are:
 - Each population from which a sample for each group is taken is assumed to be _____.
 - All samples are _____ selected and independent.
 - The populations are assumed to have _____ standard deviations (or variances).

- The factor is a _____ variable.
- The response is a _____ variable.

4.3 Null and Alternative Hypotheses

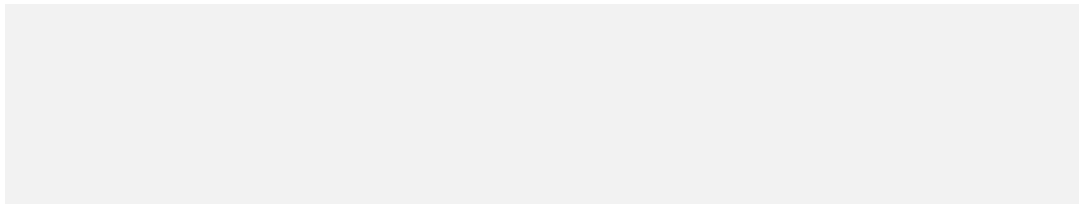
- In a one-way ANOVA, the _____ hypothesis is as follows:
 - $H_0 : \mu_1 = \mu_2 = \mu_3 \dots = \mu_k$
 - H_A : At least two of the group means $\mu_1, \mu_2, \mu_3, \dots, \mu_k$ are not equal

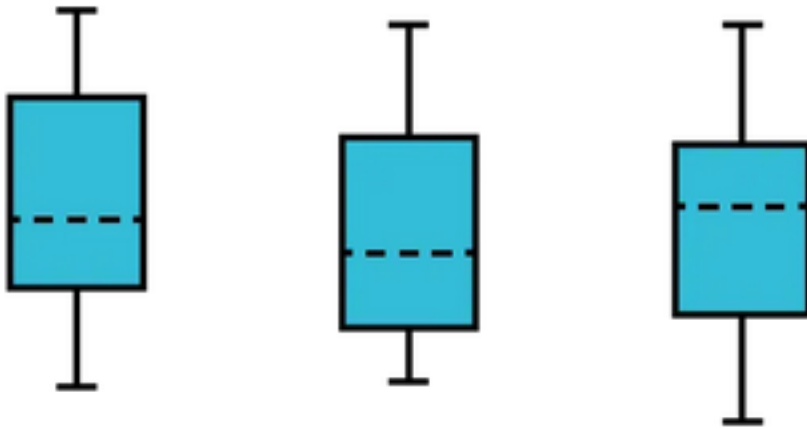
Important

Be **very** careful in how you phrase the alternative hypothesis for this test, it looks a little bit different than the other tests we've covered before!

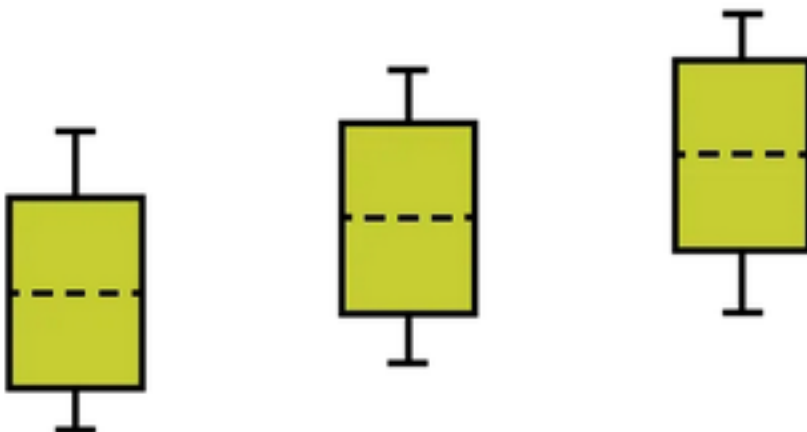
- Effectively, a statistically significant F-statistic only tells us that a difference exists _____, but not where that difference lies
 - More explanation on this in the [Extensions](#) section!

Discuss: What is the name of type of the following plots?





(a)



(b)

4.4 Extensions

- Of course, this is only the tip of the iceberg
- One question that the ANOVA alone answer: which of the groups are different?
 - If we have a one-way ANOVA test, then we can follow up with post-hoc testing, which can tell us which exactly are different than one another

- There is also the **two-way ANOVA**, which allows us to compare two or more _____ variables and their combined and interacting impact on a single dependent variable in two-way Analysis of Variance.
 - While beyond the scope of this class, _____ are an incredibly important part of a lot of more complicated studies
- Finally, there is the **repeated-measures ANOVA**

 Discuss: Which of the t-tests also dealt with comparing two sets of measurements, taken at two separate time points, on the same sample?

4.5 Effect Size for One-way ANOVA

- There are two measures of **effect size**: Eta-squared and Omega-squared

4.5.1 Eta-squared

- The first effect size estimate is **eta-squared**, or η^2
 - A measure of effect size for one-way ANOVA to describe _____ of variance in the dependent variable explained by the independent variable in this sample

$$\eta^2 = \frac{SS_{Between}}{SS_{Total}}$$

 Discuss: What other effect size coefficient is also described as 'variance explained in one variable by another'. Hint: think of comparisons between two continuous numeric variables

4.5.2 Omega-squared

- The second effect size estimate is **omega-squared** or ω^2
 - A measure of effect size for one-way ANOVA to describe proportion of variance in the dependent variable explained by independent variable in the target _____; _____ effect size estimate

$$\omega^2 = \frac{SS_{Between} - (k - 1)(MS_{within})}{SS_{total} + MS_{within}}$$

Important

Because omega-squared tries to speak to the *population*, I generally prefer it as an effect size indicator over eta-squared

5 Mean Comparisons in One-way ANOVA

5.1 Introduction

- One of the core _____ of an ANOVA test, is that it doesn't provide any information by itself on _____ groups are different than one another
 - To do these comparisons, we use **contrast/comparison tests**

- Using these, we can test _____ hypotheses about differences between our 3 or more groups
- However, you should only pursue doing mean comparisons *if* your overall one-way ANOVA is statistically significant.

! Important

Think about what our null hypothesis is under ANOVA. Very limited, right? Contrasts and comparisons are how we can move past that

5.1.1 Types of Contrasts

- We have two types of contrasts:
 - A Priori / Planned / Orthogonal Contrasts - Planned _____ of time
 - A Posteriori / Post-hoc - Performed _____ ANOVA
- Both of these types of contrasts tests make the same assumptions as the ANOVA itself

! Important

Planned comparisons are more powerful (less stringent) than post hoc comparisons which are typically more conservative. For this reasons, I'd usually say it's best to try to make predictions prior to analyzing the data.

5.1.2 Concept of Weights

- For contrast tests, **weights** are _____ to means (weighted means) to reflect the _____.
 - The means assumed to have _____ values are assigned the _____ weights
- The weights are somewhat _____, but they have to satisfy certain conditions:
 - The weights used in a comparison must sum to _____ - put another way, they are all _____ to one another.
 - Every mean _____ to have a weight even if it is zero
 - Larger weights represent _____ hypothesized means, and means that are hypothesized to be equal should be assigned the _____ weights

- Example: All of the following weights are for the same hypothesis where groups 2 and 3 are equal, and group 1 is more than each of them

G1	G2	G3
2	-1.0	-1.0
1	-0.5	-0.5
4	-2.0	-2.0

! Important

With all of this said, contrast testing, just like normal inferential testing, should never be careless!

📢 Discuss: Write a set of weight that would suffice the conditions that G1 is greater than G2 is greater than G3.

6 Conclusion

6.1 Recap

- We can think of a one-way ANOVA as being similar to an independent-samples t-test, but for more than 2 groups. Just note the differences in the null/alternative hypotheses setup
- There are some important extensions and other applications of the ANOVA that we did not comprehensively cover here. However, they do still employ the F-distribution and have the underlying focus on comparing variances to make conclusions about differences between groups. We will cover those later on!
- The F-distribution adds to the other practical distributions employed as part of hypothesis testing, such as the t-distribution and chi-squared distribution (which you learned about in EDPS-641). Just like those, it allows us to conclude whether a null hypothesis can be rejected or retained.

- On top of performing a One-way ANOVA, we can also follow it up with various Mean Comparisons in One-way ANOVA! These help us determine where our differences exist between our groups, and can offer important insights on top of our one-way ANOVA results. We continue working through these in next lecture.

Key Terms

A

Analysis of Variance (ANOVA) A family of statistical tests used to compare variances as a way to detect differences between more than two groups [2](#)

C

contrast/comparison tests A method to compare means against means in ANOVA [13](#)

E

effect size A measure of the magnitude of an effect [12](#)

eta-squared A measure of effect size for one-way ANOVA to describe proportion of variance in the dependent variable explained by the independent variable in this sample [12](#)

F

F-distribution A distribution developed by Ronald Fisher, expressed as a ratio, used commonly for the ANOVA family of tests [3](#)

F-ratio Another name for the F-statistic derived from an ANOVA test, based upon the ratio of the variances between the samples to the variances within the samples [5](#)

M

mean square The computational representation of variation between groups [6](#)

O

omega-squared A measure of effect size for one-way ANOVA to describe proportion of variance in the dependent variable explained by independent variable in the target population; unbiased effect size estimate [13](#)

P

post-hoc testing Statistical testing done after a significant ANOVA to tease apart where exactly differences lie between the groups [11](#)

R

repeated-measures ANOVA A type of ANOVA used when measurements are taken on the same sample 3 or more times, to compare against each other at different time points 12

T

two-way ANOVA A type of ANOVA that allows for testing two independent (grouping) variables impact on a single dependent (outcome) variable 12

V

Variances between samples How much differences there are between the 3 or more groups that we are comparing 6

Variances within samples How much variances occur within each of the groups we are comparing against one another 6

W

weights Relative amounts assigned to describe the relationships between variables in contrast testing 14

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms